Honours Linear Algebra - Week 1

7 and 9 September 2026

7 Sept. / Linear equations

$A = (a_{ij}) \in M_{m\times n}(\mathbb{R})$: m rows, n columns. Column vectors throughout.

$$\sum_{j=1}^{n} a_{ij}x_j = b_i \quad \longleftrightarrow \quad Ax = b, \qquad [A \mid b] \in M_{m\times(n+1)}.$$

Notation.

$$(A^T)_{ij} = a_{ji}, \quad A^T \in M_{n\times m}, \qquad I_n = (\delta_{ij}),$$
$$(A+B)_{ij} = a_{ij} + b_{ij}, \qquad (cA)_{ij} = ca_{ij}.$$

Upper triangular: $a_{ij} = 0$ for $i > j$; lower: $i < j$; diagonal: $i \neq j$.

Row operations.

$$R_i \leftrightarrow R_j, \qquad R_i \leftarrow cR_i \ (c \neq 0), \qquad R_i \leftarrow R_i + cR_j \ (i \neq j).$$

All reversible $\Rightarrow$ same solutions. Operate on the right-hand side too.

REF. Zero rows last; leading 1's move right down the rows. RREF: pivot columns zero elsewhere. This lecture takes pivots to be 1.

Solving. Eliminate $\rightarrow$ back substitute. For n variables and r coefficient pivots:

$[0 \ \cdots \ 0 \mid 1]$ occurs	no solution
consistent, $r = n$	unique
consistent, $r < n$	$n - r$ free variables

Ex. Nonzero rows of the echelon system:

$$x_1 - x_2 + 4x_3 + 5x_4 + 4x_5 = 0,$$
$$x_3 + 2x_4 + 6x_5 = 1, \qquad x_5 = 3.$$

$x_2 = s$, $x_4 = t \Rightarrow x_3 = -17 - 2t$, $x_1 = 56 + s + 3t$.

$$x = \begin{pmatrix} 56 \\ 0 \\ -17 \\ 0 \\ 3 \end{pmatrix} + s\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} 3 \\ 0 \\ -2 \\ 1 \\ 0 \end{pmatrix}, \qquad s, t \in \mathbb{R}.$$

Ex. $3x_1 - 9x_2 = 4$, $-2x_1 + 6x_2 = h \Rightarrow 0 = h + 8/3$. Consistent iff $h = -8/3$; then $(x_1, x_2) = (4/3 + 3t, t)$.

Theorem. Homogeneous, $n > m \Rightarrow$ nonzero solutions:

$$x = 0 \text{ solves it}, \qquad n - r \geq n - m > 0.$$

At least one free variable. Square RREF with n pivots $= I_n$.

9 Sept. (I) / Inverse and LU

$(AB)_{ij} = \sum_k a_{ik}b_{kj}$; $(m \times n)(n \times p) \to m \times p$.

Inverse. $AA^{-1} = A^{-1}A = I$; square A only.

$$A \text{ invertible} \iff Ax = b \text{ has exactly one solution for every } b.$$

Row-reduce $[A \mid I] \to [Q \mid E]$: $Q = I \Rightarrow A^{-1} = E$; otherwise no inverse.

Why: $E = E_k \cdots E_1$ (elementary matrices), $EA = I$. Each E_i invertible $\Rightarrow A = E^{-1} \Rightarrow AE = I$ too.

$$(AB)^{-1} = B^{-1}A^{-1}, \qquad (A^T)^{-1} = (A^{-1})^T,$$
$$(B^{-1}A^{-1})(AB) = I, \qquad (A^{-1})^T A^T = (AA^{-1})^T = I.$$

Other product orders similarly. Inverses reverse order.

Theorem. Triangular + nonzero diagonal $\Rightarrow$ inverse triangular of the same type.

Upper case: solve $Ax = e_j$ backwards. $i > j \Rightarrow x_i = 0$; division only by $a_{ii} \neq 0$. These x's form A^{-1}.

LU. Elimination without row swaps:

$$E_k \cdots E_1 A = U \quad \Rightarrow \quad A = LU, \quad L = E_1^{-1} \cdots E_k^{-1}.$$

L: unit lower triangular; elimination multipliers below the diagonal.

Ex. Opening matrix from *The Nine Chapters*:

$$A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}, \qquad \begin{aligned} R_2 &\leftarrow R_2 - \tfrac{2}{3}R_1, \\ R_3 &\leftarrow R_3 - \tfrac{1}{3}R_1, \\ R_3 &\leftarrow R_3 - \tfrac{4}{5}R_2. \end{aligned}$$

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 2/3 & 1 & 0 \\ 1/3 & 4/5 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} 3 & 2 & 1 \\ 0 & 5/3 & 1/3 \\ 0 & 0 & 12/5 \end{pmatrix}}_{U}.$$

$Ax = b$: first $Ld = b$ (forwards), then $Ux = d$ (backwards).

$$b = \begin{pmatrix} 39 \\ 34 \\ 26 \end{pmatrix} \Rightarrow d = \begin{pmatrix} 39 \\ 8 \\ 33/5 \end{pmatrix} \Rightarrow x = \begin{pmatrix} 37/4 \\ 17/4 \\ 11/4 \end{pmatrix}.$$

With swaps: $PA = LU$, so $Ld = Pb$, $Ux = d$. Slides' $A = PLU$ uses the inverse permutation.

Factor once $O(n^3)$; each new b: $O(n^2)$. Reuse L, U rather than form A^{-1}.

9 Sept. (II) / Fields; vector spaces

Def.1.1. Field K: $(K,+)$, $(K\setminus\{0\},\cdot)$ abelian groups + distributivity; $0\neq 1$.

Examples: $\mathbb{Q},\mathbb{R},\mathbb{C}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{R}(t)$, $\mathbb{F}_p$ (p prime). $\mathbb{Z}/6\mathbb{Z}$ fails: $2\cdot 3=0$.

Def.1.4. Vector space over K: addition $V\times V\to V$, scalar multiplication $K\times V\to V$; for $u,v,w\in V$, $c,d\in K$,

$$\begin{aligned} u+v&=v+u, & (u+v)+w&=u+(v+w),\\ u+0&=u, & u+(-u)&=0,\\ (cd)u&=c(du), & 1u&=u,\\ c(u+v)&=cu+cv, & (c+d)u&=cu+du. \end{aligned}$$

$0\in V$ and negatives must exist. Scalars $\in K$; vectors $\in V$.

Prop.1.5–7. Unique zero, unique negative; cancellation; $cu=0 \iff c=0$ or $u=0$.

$$\begin{aligned} 0'&=0'+0=0,\\ u',u'' \text{ negatives of } u:\quad u'&=u'+(u+u'')=(u'+u)+u''=u'',\\ 0u&=(0+0)u=0u+0u\Rightarrow 0u=0,\\ c0&=c(0+0)=c0+c0\Rightarrow c0=0,\\ cu&=0,\ c\neq 0\Rightarrow u=c^{-1}(cu)=0. \end{aligned}$$

Cancellation: add $-u$. Also $u+(-1)u=(1-1)u=0\Rightarrow(-1)u=-u$.

Examples. $\{0\}$, K^n, $M_{m\times n}(K)$, $K[t]$. Over $\mathbb{R}$: $C^0(\mathbb{R})$; $\{f: f''+f=0\}$, since

$$(af+bg)''+(af+bg)=a(f''+f)+b(g''+g)=0.$$

$\mathbb{C}$ over $\mathbb{R}$: yes. $\mathbb{R}$ over $\mathbb{C}$: no, $i\cdot 1\notin\mathbb{R}$.

Non-examples (usual operations).

- $x,y\geq 0$: multiplying $(1,0)$ by -1 leaves the set.
- Invertible matrices; degree exactly n; $f(0)=1$: zero missing.
- A line not through 0: same obstruction.

Subspaces; span

Definition 1.9. A subset $U \subset V$ is called a *subspace* of V if U is itself a vector space.

To check whether a subset $U \subset V$ is a subspace, we only need to check zero and closures.

Proposition 1.10. Let V be a vector space. A subset $U \subset V$ is a subspace of V iff it satisfies all the following conditions:

(1) Zero exists: $\mathbf{0} \in U$.
(2) Closure under addition: $\mathbf{u}, \mathbf{v} \in U \Longrightarrow \mathbf{u} + \mathbf{v} \in U$.
(3) Closure under multiplication: $\mathbf{u} \in U, c \in K \Longrightarrow c \cdot \mathbf{u} \in U$.

Note. By Proposition 1.7 and (3) (taking $c = 0$), (1) can be replaced by "U is non-empty".

9 Sept., vector spaces, p.13.

Test. $U \neq \varnothing$ and $au + bv \in U$ for all $u, v \in U$, $a, b \in K$. Identities inherited from V; $-u = (-1)u \in U$.

Examples: $x + y + z = 0$ in $\mathbb{R}^3$; $K_n[t]$ (degree $\leq n$, including 0); $\{f : f(0) = 0\}$. $\mathbb{R}^2$ itself is not a subset of $\mathbb{R}^3$; $\{(x, y, 0)^T\}$ is.

Null space. $A \in M_{m\times n}(K)$:

$$\mathrm{Nul}(A) = \{x \in K^n : Ax = 0\} \leq K^n, \qquad A(au + bv) = aAu + bAv = 0.$$

Def.1.11. $\mathrm{Span}(S) = \{\text{finite linear combinations of vectors in } S\}$; $\mathrm{Span}(\varnothing) = \{0\}$.

Thm.1.12. $\mathrm{Span}(S)$ is a subspace, the smallest one containing S. In particular $\mathrm{Span}(U) = U$ for a subspace U.

Proof: add/scale combinations; still finite combinations. Any subspace containing S contains all of them.

$$u_i = \sum_j a_{ij} v_j \quad \Rightarrow \quad \sum_i c_i u_i = \sum_j \left(\sum_i c_i a_{ij} \right) v_j.$$

Ex.1.27.

$$\begin{pmatrix} a - 3b \\ b - a \\ a \\ b \end{pmatrix} = a \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} -3 \\ 1 \\ 0 \\ 1 \end{pmatrix}.$$

$a, b \in \mathbb{R}$ arbitrary $\Rightarrow U = \mathrm{Span}\{(1, -1, 1, 0)^T, (-3, 1, 0, 1)^T\} \leq \mathbb{R}^4$.

Back to equations. If $Ax_p = b$,

$$Ax = b \iff A(x - x_p) = 0 \iff x \in x_p + \mathrm{Nul}(A).$$

Homogeneous: subspace. Inhomogeneous: translate; $b \neq 0 \Rightarrow 0$ not a solution.